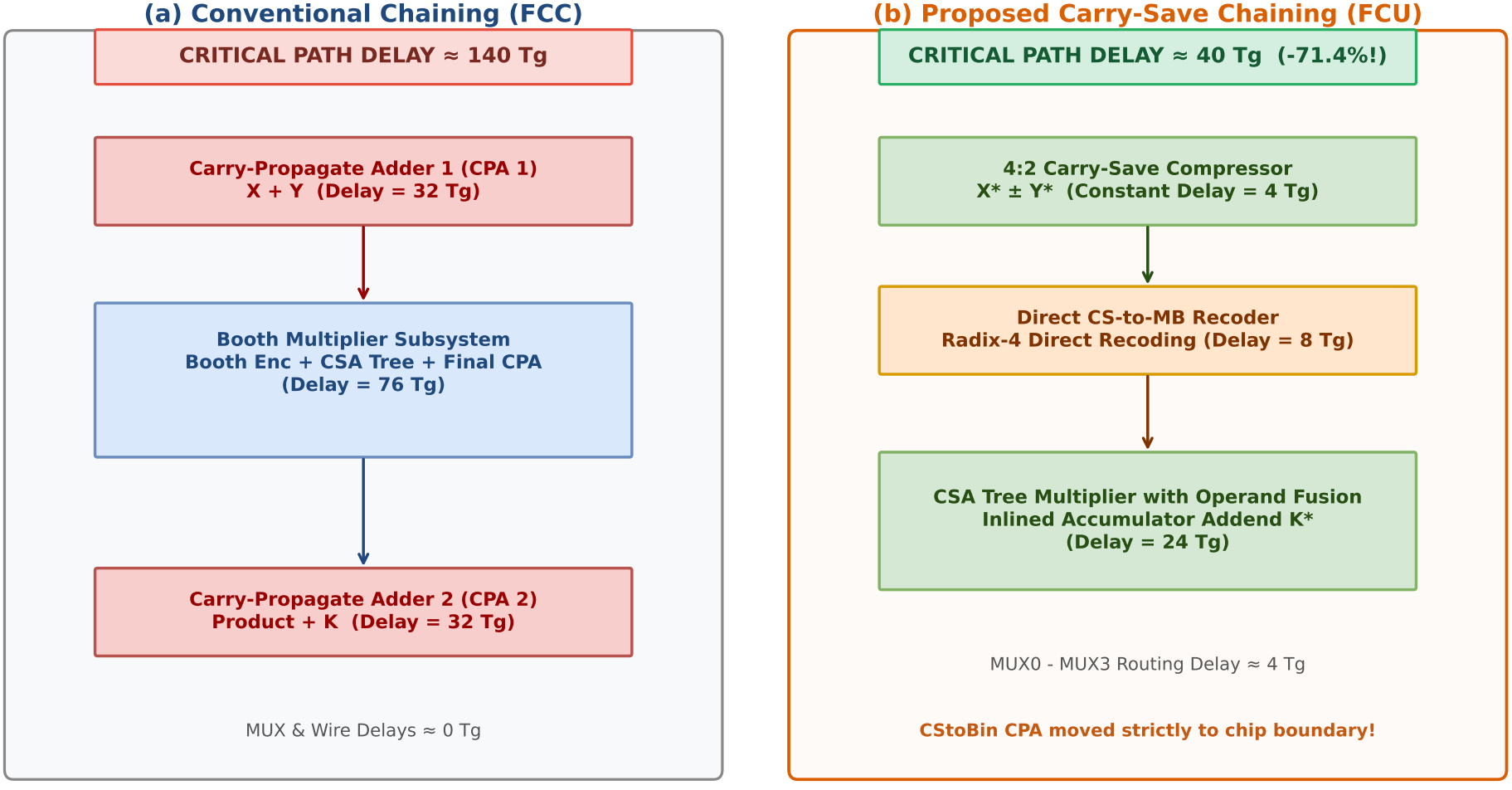
(a) Conventional Chaining (FCC)
CRITICAL PATH DELAY ≈ 140 Tg
Carry-Propagate Adder 1 (CPA 1)
X + Y (Delay = 32 Tg)
Booth Multiplier Subsystem
Booth Enc + CSA Tree + Final CPA
(Delay = 76 Tg)
Carry-Propagate Adder 2 (CPA 2)
Product + K (Delay = 32 Tg)
MUX & Wire Delays ≈ 0 Tg
(b) Proposed Carry-Save Chaining (FCU)
CRITICAL PATH DELAY ≈ 40 Tg (-71.4%!)
4:2 Carry-Save Compressor
X* ± Y* (Constant Delay = 4 Tg)
Direct CS-to-MB Recoder
Radix-4 Direct Recoding (Delay = 8 Tg)
CSA Tree Multiplier with Operand Fusion
Inlined Accumulator Addend K*
(Delay = 24 Tg)
MUX0 - MUX3 Routing Delay ≈ 4 Tg
CStoBin CPA moved strictly to chip boundary!